

ISO 2768 general tolerances: f, m, c, v and H, K, L tables

1. Linear dimensions (ISO 2768-1)

Permissible deviations in millimetres, by nominal size range (over ... up to and including ...). A dash means the standard gives no value: the dimension must carry its own tolerance.

Class	0.5 - 3	3 - 6	6 - 30	30 - 120	120 - 400	400 - 1000	1000 - 2000	2000 - 4000
f (fine)	± 0.05	± 0.05	± 0.1	± 0.15	± 0.2	± 0.3	± 0.5	—
m (medium)	± 0.1	± 0.1	± 0.2	± 0.3	± 0.5	± 0.8	± 1.2	± 2
c (coarse)	± 0.2	± 0.3	± 0.5	± 0.8	± 1.2	± 2	± 3	± 4
v (very coarse)	—	± 0.5	± 1	± 1.5	± 2.5	± 4	± 6	± 8

Below 0.5 mm, the standard doesn't apply: such dimensions must always carry their own tolerance.

2. Radii, chamfers and angles

External radii and chamfer heights (± mm):

Class	0.5 - 3	3 - 6	> 6
f (fine)	± 0.2	± 0.5	± 1
m (medium)	± 0.2	± 0.5	± 1
c (coarse)	± 0.4	± 1	± 2
v (very coarse)	± 0.4	± 1	± 2

Angles, by length of the shorter side of the angle (mm):

Class	≤ 10	10 - 50	50 - 120	120 - 400	> 400
f (fine)	±1°	±0°30'	±0°20'	±0°10'	±0°5'
m (medium)	±1°	±0°30'	±0°20'	±0°10'	±0°5'
c (coarse)	±1°30'	±1°	±0°30'	±0°15'	±0°10'
v (very coarse)	±3°	±2°	±1°	±0°30'	±0°20'

3. General geometrical tolerances H, K, L (ISO 2768-2)

Straightness and flatness, by nominal length:

Class	≤ 10	10 - 30	30 - 100	100 - 300	300 - 1000	1000 - 3000
H	0.02	0.05	0.1	0.2	0.3	0.4
K	0.05	0.1	0.2	0.4	0.6	0.8
L	0.1	0.2	0.4	0.8	1.2	1.6

Perpendicularity, by length of the shorter side:

Class	≤ 100	100 - 300	300 - 1000	1000 - 3000
H	0.2	0.3	0.4	0.5
K	0.4	0.6	0.8	1
L	0.6	1	1.5	2

Symmetry, by length:

Class	≤ 100	100 - 300	300 - 1000	1000 - 3000
H	0.5	0.5	0.5	0.5
K	0.6	0.6	0.8	1
L	0.6	1	1.5	2

Circular run-out (radial, axial): 0.1 mm in H, 0.2 mm in K, 0.5 mm in L.

Reference values taken from the article <https://www.baouch.fr/en/blog/tolerances-generales-iso-2768/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.